

```
SELECT param_name, step,
       COUNT(DISTINCT cksum) AS n
FROM trace
GROUP BY param_name, step
HAVING n > 1;
```

*no topology filter:
every TP-sharded tensor
becomes a violation*



(a) π_{schema} only (83.3 FP/1M)

```
SELECT param_name, step,
       COUNT(DISTINCT cksum) AS n
FROM trace
WHERE tensor_model_
      parallel = FALSE
GROUP BY param_name, step
HAVING n > 1;
```

*guard in the query:
sharded tensors filtered
before the check*

*still fires at step 0,
before weight sync*



(b) $+\pi_{\text{topo}}$ (shards excluded)

```
SELECT param_name, step,
       COUNT(DISTINCT cksum) AS n
FROM trace
WHERE tensor_model_
      parallel = FALSE
AND stage =
   'after_optimizer'
GROUP BY param_name, step
HAVING n > 1;
```

*valid only after
the optimizer step:
clean run silent,
bug DETECTED*



(c) verified rule (25.8 FP/1M, 0/17 FP)